MIL-C-60184(MU)
AMENDMENT 5
12 July 1968
SUPERSEDING
AMENDMENT 4
25 August 1967

MILITARY SPECIFICATION

COMPUTER SET, GUN DIRECTION: 10525735

This amendment forms a part of Military Specification MIL-C-60184(MU), dated 18 March 1965.

Paragraph 3.5.2: Delete the first sentence and substitute the following:

"The computer while in its field case, shall meet the requirements of 3.7.1 (The only tests to be performed utilize connector J013 to perform the output voltage measurements of Table I at a nominal 120 VAC line to neutral input voltage and the kickout tests, 4.5.3(d), of Table VIII), and 3.7.3 through 3.7.9. These tests are to be performed at an ambient temperature of 75 ± 15°F while the computer is subjected for 6 hours to an atmosphere of sand and dust."

Paragraph 3.5.3.1b: Delete and substitute:

"b. The temperature warning indicator lamp shall energize at an ambient temperature greater than 117°F or an internal temperature lower than -18°F."

Paragraph 3.5.3.1c: Delete and substitute:

"c. The computer shall de-energize at an ambient temperature greater than 125°F or an internal temperature lower than -40°F."

Add the following new subparagraph to 3.5.3.1:

"d. The computer may de-energize after sustained operation (sustained operation is defined as, operation of at least 1 hour) at an ambient temperature of +125°F. This is not to be considered a failure. In order to achieve sustained operation after this occurrence, the chamber temperature will be reduced, in increments of 2°F, to the point where operation may be sustained, but not less than +121°F."

* Paragraph 3.7.1:

(a) Change "132 volts rms and above" to "132 ± 4 volts rms".

(b) Change "108 volts rms and above" to "106 ± 4 volts rms".

(c) Change "140 volts rms and above" to "140 ± 4 volts rms".

FSC 1220

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(d) Change "100 volts rms and above" to "98 ± 4 volts rms".

(e) Change "80 volts rms and above" to "82 ± 4 volts rms".

Add the following subparagraph to requirement 3.7.1:

"e. The power ready indicator which illuminates when the computer is energized shall blink continuously when the blower cut-off switch is actuated manually and the marginal test switch is in the "OFF" position. This condition shall also exist when the rear cover is installed as the cover automatically actuates the blower cut-off switch."

Delete the second sentence of 3.7.7 and substitute the following:

"The delay shall be minimized such that on either (a) rapid entry (Maximum of 0.25 second time interval between key depressions) of any digit on the keyboard, the nixie indicator shall light before the next depression of a key or (b) using a routine which causes the nixie display of a character or characters to shift at a rate equivalent to keyboard entry (maximum of 0.25 second time interval between shifts of nixie display), the display shall shift at a minimum rate of 4 nixie positions per second."

Paragraph 3.8: Change references to "70 hour run-in" to "50 hour run-in"; delete "and 3.8.1" in second sentence and delete third sentence.

Paragraph 4.3.1: Delete and substitute:

"4.3.1 General.- One unit shall be selected at random as a standard sample from each 50 produced. The sample shall meet the requirements and tests in Table B and paragraph 4.2.1.1(a) through 4.2.1.1(l)."

Delete the third sentence of 4.5.1.2 and substitute the following:

"During the operating portion of the test, defined in 4.5.2, 4.5.3 (only tests of 4.5.3 to be performed are those voltage measurements of Table 1, whose outputs are available at connector J013, using a nominal 120 VAC line to neutral input voltage and the kickout tests of Table VIII, outputs also available at J013), and 4.5.5 to 4.5.11, inclusive, the computer air filters shall be replaced as required to obtain proper operation.

Paragraph 4.5.1.6: Delete in its entirety and substitute:

"4.5.1.6 Salt spray.- This test shall be conducted in accordance with the salt spray test defined in Method 101, Condition B of MIL-STD-202 using a 5 percent salt solution. During exposure, the computer rear field case cover will be installed and the front cover removed at the end of the exposure period (48 hours). Salt deposits shall be removed from all exposed surfaces by a gentle wash, not exceeding 30 minutes, in running water not warmer than 100°F, using a soft-hair brush or plastic-bristle brush. The

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computer shall then be air dried in a controlled environment at 72°F ± 5°F and a relative humidity of 30 to 50 percent for a period of 24 hours. Subsequent to the salt spray test, the computer shall be visually examined for physical damage then tested as outlined in 4.5.5 through 4.5.11 inclusive."

Paragraph 4.5.1.8, first sentence: Delete "with all covers in place.". Second sentence: Delete "of the front cover". Fourth sentence: Delete "covers shall be removed".

Paragraph 4.5.3(d): Delete "+.5 ± .5 volts" and substitute "0 to +2 volts".

Paragraph 4.5.3.1: Add new subparagraph:

"(c) Energize computer and observe that when the power ready light comes on it shall not blink. Manually actuate the blower cut-off switch to cause the power ready light to blink continuously. If the blower cut-off switch is actuated while the computer is energized, the power ready light may be de-energized indicating the computer has entered on out of sync mode. This is considered a normal condition and is not to be considered a computer malfunction."

Paragraph 4.5.4: Add the following:

"A maximum of 20 readings checked during the following tests of this paragraph may fall outside the specified limits within the limitations listed below:

a.	4.5.4.3(f)	65 ± 10%
b.	4.5.4.4(e)	8.6 volts
c.	4.5.4.5(d)	.55 volts
d.	4.5.4.6(d)	.55 volts"

Paragraph 4.5.4.3(f), last sentence: Delete "75" and substitute "65".

Paragraph 4.5.4.3(h): Delete "1%" and substitute "5%"

Paragraph 4.5.9.5(b): Delete in its entirety and substitute:

"(b) The signal at Pin W on J10 shall be equivalent to the signal shown in Figure 2(c). The information signal for a binary "1" shown in Figure 2(b), shows the correspondence between the information signal and the strobe signal. Information which appers on the data lines at times when the strobe is not true (true is the -4.0 ± 2.0 volt level shown in Figure 2) is to be disregarded."

Paragraph 4.5.9.5: Add subparagraphs (c) and (d):

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"(c) Generate and execute a routine to input field data code."

"(d) The signal at Pin Y on J17 shall be equivalent to the signal shown in Figure 2(a)."

Paragraph 4.5.9.9: Delete the second and third sentences and substitute the following:

"The routine shall either (a) cause the nixie indicators to display each digit entered from the keyboard before the depression of the next key when entry is made at a minimum rate of 4 digits per second, or (b) cause the nixie display of a character or characters to shift at a rate equivalent to keyboard entry (minimum shift rate of 4 nixie positions per second) for compliance with 3.7.7."

Paragraph 4.5.10(a): Add "Rev A" after "8213315-17".

Paragraph 4.5.10(b): Change "9 00000 00000 05055" to 9 00000 00001 05055."

Paragraph 4.5.10.1(e): Delete in its entirety and substitute the following:

"(e) Depress set-up button. Using targets 1 through 5 of Table XXXI, enter the problems of Table XXXII.

Enter:

Battery A	in	Marginal 5
Battery B	in	Marginal 4
Battery C	in	Marginal 3
Battery D	in	Marginal 2
Battery E	in	Marginal 1

Select matrix position H-G. Depress SM button and then depress "0". Obtain appropriate solutions of Table XXXIII."

Delete Paragraphs "4.5.12 through 4.5.12.4" and substitute:

"4.5.12 Run-in.- As specified in 3.8 and after each computer has successfully passed all the performance tests, each computer shall be run for a total of 50 hours exclusing down-time. The tests shall be run for at least 4 hours at -22°F, 4 hours at +125°F and at least 30 hours after the -22°F, +125°F tests are completed."

Change "4.5.12.5 and 4.5.12.6" to "4.5.12.1 and 4.5.12.2" respectively.

* Table I, line 14: Delete"-25.0 ± 1.08*" and substitute "-26.5 ± 1.0*". Line 15: Delete "-30.5 ± 1.28*" and substitute "-30.5 ± 1.83*".

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* Table IX, Column A: Delete and substitute the following:

"A Line Voltage ±4 volts rms
140
98
82
106
132"

Figure 1: Delete lower limit range "-5.4v" to "-6.6v" and substitute "-3.7 ± 0.8v".

Figure 2, Note A: Delete "not less than 2 microsec"; Note B: Delete "not less than 2 microsec" and substitute "not more than 2 microseconds"; Note C: Delete "nor more than 2 microsec" and substitute "nor more than 3 microseconds".

Table XXXIII: Delete and substitute:

TABLE XXXIII

PROBLEM SOLUTIONS
MET. STANDARD

Bat.	Target	Deflection Low	Deflection High	Fuse Time Low	Fuse Time High	Quad. Elev. Low	Quad. Elev. High	
A	1	1500	1718	4.4	38.9	170	1412	NSL
A	6	94	116	20.0	29.6	610	958	
B	2	2986	3138	7.1	46.0	158	1447	NSL
B	7			...0	(Indicates Out of Range)			
C	3	2094	2194	11.9	36.8	220	1270	
C	8			...	(Indicates Out of Range)			
D	4	1882	1982	8.0	38.0	204	1380	NSL
D	9			...0	(Indicates Out of Range)			
E	5	587	756	8.6	42.4	136	1373	NSL
E	10	67	103	19.8	34.9	490	1047	

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Table XXXIV: Delete and substitute:

TABLE XXXIV

PROBLEM SOLUTIONS
MET "H" TAPE

Bat.	Target	Deflection Low	Deflection High	Fuse Time Low	Fuse Time High	Quad. Elev. Low	Quad. Elev. High	
A	1	1499	1706	4.4	38.9	170	1408	NSL
A	6	99	121	20.4	29.5	623	954	
B	2	2985	3121	7.1	46.0	158	1448	NSL
B	7			...0	(Indicates Out of Range)			
C	3	2093	2181	11.8	36.9	217	1272	NSL
C	8	1753	1759	25.1	28.9	694	819	
D	4	1881	1968	8.0	38.0	205	1378	NSL
D	9			...0	(Indicates Out of Range)			
E	5	587	767	8.6	42.4	137	1371	NSL
E	10	71	110	20.1	34.9	498	1048	

Paragraph 30.3.1: Delete second sentence and substitute:

"Information is loaded in octal Teletype or Field data code."

Paragraph 30.3.1(c): Insert the following sentence between the first and second sentence:

"In initial program fill the last character entered is normally a HALT code". Change last sentence to read: "then the COMPUTE code will start computation at the address specified in L when the computer leaves the manual HALT mode.

Paragraph 30.3.2.1(c)(6): Delete "Manual" and substitute "Program".

Paragraph 30.3.2.3(a): Add "by character".

Paragraph 30.3.2.3(b): Add "by word".

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Paragraph 30.3.2.3(c): Add "by character".

Paragraph 30.3.2.3(d): Add "by word".

Paragraph 30.4.1: Delete "ENTER" and substitute "INITIATE".

Paragraph 30.4.2.3(a): Delete "8-bit Character" and substitute "Alphanumeric".

*The outside margins have been marked to indicate where additions to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notation. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notation and relationship to the last pervious amendment.

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